

Global Memory

Korea Memory Export Tracker (Aug): Samsung remained ahead of SK hynix in HBM momentum



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We track the Korea export data as it is a good early indicator for Samsung's & SK hynix's HBM revenue in the same quarter, and update this tracker with the Aug data. Details of our methodology can be found in [our prior note](#). The dataset can be downloaded at this [link](#).

Overall HBM export was solid, up 17% QoQ vs. May. It rebounded seasonally by 19% MoM. QoQ growth was 17% QoQ. YoY increase was also solid at 77%.

Data continued to support very strong 3Q26 HBM revenue for Samsung. Export from S. Chungcheong (where Samsung packages HBM) was up 61% QoQ vs. May. And regression based on Jul+Aug data predicts Samsung's HBM revenue to rise 72% QoQ in 3Q26, 23% ahead of our forecast.

No major improvement for the data associated with SK hynix. Export from N. Chungcheong & Gyeonggi was up mildly by 7% MoM but still down 21% QoQ. Regression still shows HBM revenue to drop 14% QoQ in 3Q26, significantly below our projection. We think this was likely due to the delayed HBM4 shipment and also note the ramp of HBM4, when it finally happens, may make monthly linearity unseasonal and our regression less predictive.

Data suggested a continued HBM4 ramp at Samsung. We track "value per weight" as it may be directionally suggestive of HBM price change. Aug "value per weight" rose another 8% MoM for Samsung but was still flattish for SK hynix, signaling earlier and continued HBM4 ramp by Samsung. Samsung's "value per weight" has also more than doubled from the prior level, suggesting price perhaps improving notably from HBM3/3E to 4.

Data showed HBM price is still largely insulated from the volatility of conventional memory price. Conventional memory price surged 5x since 3Q25, but HBM price, as suggested by value per weight, has been rising much more mildly.

Export to Malaysia exhibited continued strength. It rose further to US\$1.6B in Aug, or 33% of the total HBM export, with the increase from both Samsung and SK hynix. Generally export to Malaysia has been growing meaningfully over the past few quarters. We assume it's for Intel's EMIB but are puzzled why it happened so early.

Overall data continued to show solid HBM growth for Samsung but likely slower progress for SK hynix. Aug data continued to suggest Samsung HBM revenue can outgrow SK hynix in 3Q26, and may claim market share lead too. This is in line with our view of improving HBM position by Samsung. That said we think SK hynix HBM4 ramp should eventually happen this year and help gain some share back. Additionally, HBM contract price increase should benefit them & the industry broadly next year.

Conventional memory price rise should continue in 3Q26. We see continued price increase in 3Q26 & also HBM price hike in 2027. We remain structurally constructive on Samsung, SK hynix & Micron but cautious on KIOXIA on China competition (see our [sector update](#)). Our [detailed SOTP valuation](#) also finds KIOXIA over-valued.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	21 Sep 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
005930.KS (SEC- Samsung)	O	KRW	274,000	440,000	220.0%	KRW	6,611.53	48,393	77,273	41.4	5.7	3.5
005935.KS (SEC-Pref - Samsung)	O	KRW	208,000	374,000	200.2%	KRW	6,611.53	48,393	77,273	31.5	4.3	2.7
SMSN.LI (Samsung)	O	USD	4,636.00	7,350.00	226.1%	USD	116.15	812.39	1,290.80	39.9	5.7	3.6
000660.KS (SK hynix)	O	KRW	1,868,000	3,300,000	405.4%	KRW	60,341	395,677	568,862	31.0	4.7	3.3
MU (Micron)	O	USD	1,015.80	1,300.00	508.9%	USD	8.29	67.39	158.99	122.6	15.1	6.4
285A.JP (KIOXIA)	U	JPY	54,570	40,000	1080.9%	JPY	1,014.00	10,013	9,656.84	53.8	5.4	5.7
ASIA			1,953.49									
EM			1,817.41									
SPX			7,650.50									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

MU estimate is Adjusted EPS; MU valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Samsung Electronics: We rate Samsung Electronics Outperform with price target of KRW 440,000.

SK hynix : We rate SK hynix Outperform with price target of KRW 3,300,000.

Micron: We rate Micron Outperform with price target of US\$1,300.00.

KIOXIA: We rate KIOXIA Underperform with price target of JPY 40,000.

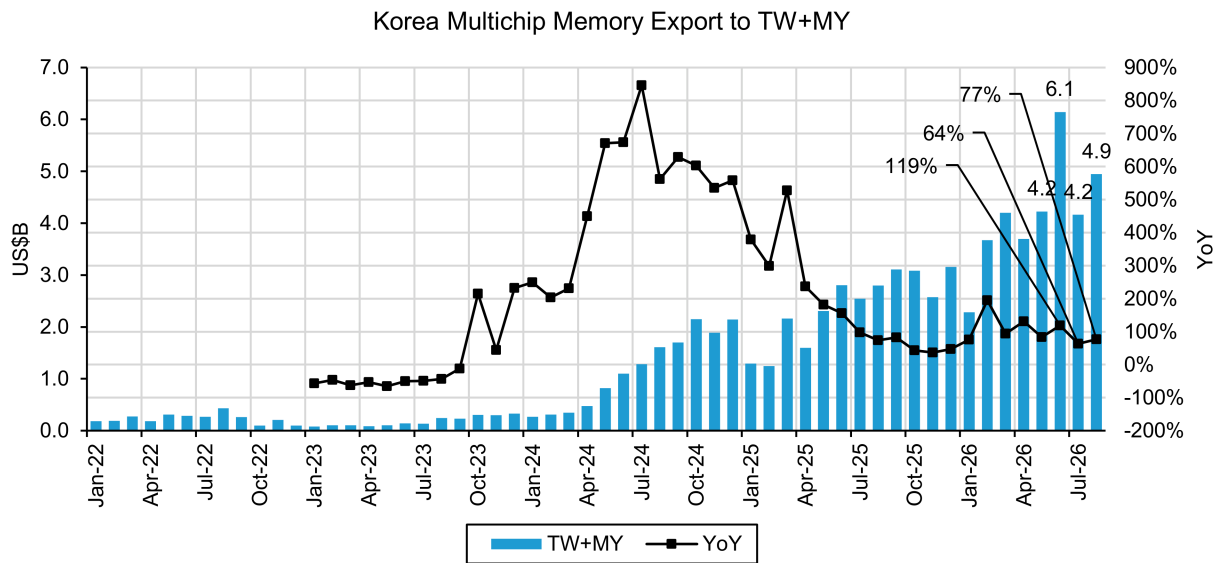
DETAILS

Korea Customs Service has released export data for Aug. As [our previous note](#) found, the export data for certain memory products bears close correlation with HBM revenues of Samsung and SK hynix. We hence track the monthly data to provide investors a preview on HBM revenues from the two companies in the same quarter. Details of our methodology can be found in [the previous note](#). You may also download the export data at this [link](#).

Aug export data continued to suggest very strong growth of Samsung HBM revenue into 3Q26, whereas export data from locations associated with hynix fabs remained lukewarm.

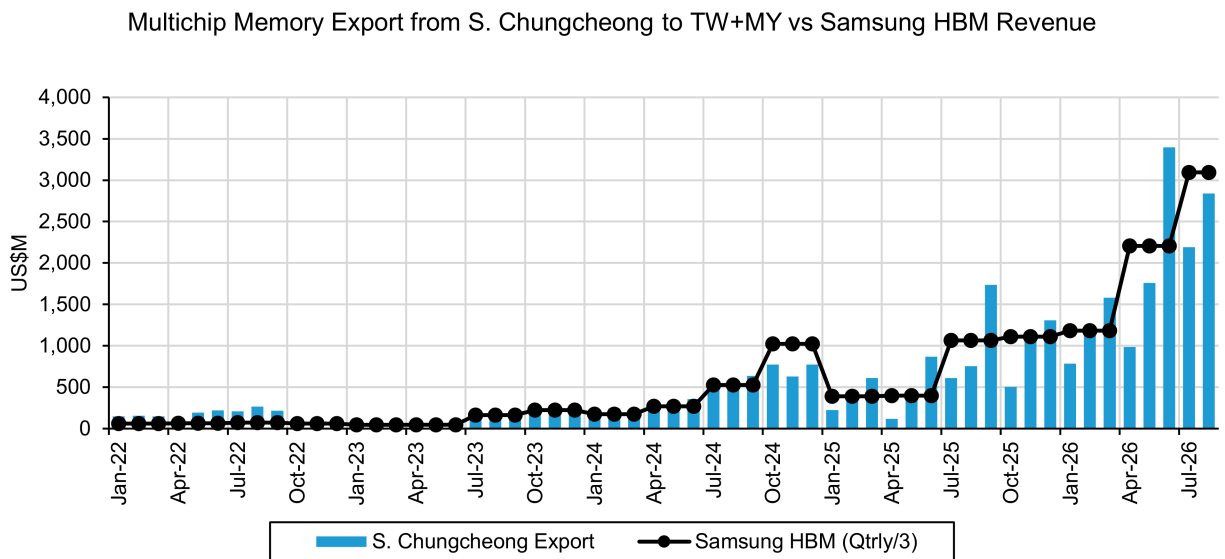
- Aug Korea total multichip memory export to Taiwan and Malaysia was up by 19% MoM, rebounding from Jul's seasonal dip. QoQ growth (vs May) was solid at 17% and YoY growth also remained strong at 77% YoY ([Exhibit 1](#)). As we analyzed before, multichip memory export to Taiwan and Malaysia tracked closely with HBM revenues of Samsung and SK hynix and therefore the Aug data indicated continued strong momentum into 3Q26.
- If we focus on the export from S. Chungcheong Province, where Samsung's back-end fabs are and likely where its HBM is packaged, Aug export recovered by 30% MoM to US\$2.8B, in line with its monthly seasonality ([Exhibit 2](#)). QoQ growth vs May was 61%, and Jul+Aug QoQ growth was 83%, indicating very strong growth for 3Q26 ([Exhibit 3](#)), and according to the regression using historical data, the Jul+Aug data suggested ~US\$11.4B in HBM revenue for Samsung in 3Q26, notably ahead of our estimate of US\$9.3B by ~23%, and represented 72% QoQ growth ([Exhibit 4](#)). We get roughly the same results if we run the regression using only data points from 1Q25 ([Exhibit 6](#)). Samsung's HBM export monthly seasonality has consistently been very backend loaded since 1Q25, with first month of quarter averaging less than 20% of quarterly total. So far 3Q26 appears to be following a similar pattern and we expect to see further sequential growth in Sep ([Exhibit 5](#)). Overall Samsung's HBM export has been stronger than our expectation in recent months, and does align with Samsung's guidance of the rapid ramp of HBM4 sales in 2H26. We hence remain positive on Samsung's HBM share gain this year (see Samsung 2Q26 earnings [takeaway](#)).
- For SK hynix, we have identified that its HBM export should come nearly all from N. Chungcheong Province and Icheon City, where its wafer fabs are. However since Sep 1, Korea Customs has ceased to disclose by item trade data down to 10 digit HS code at city level, and hence we can no longer obtain multichip memory export data from Icheon City. However we've found that multichip memory export to Taiwan & Malaysia from Gyeonggi Province, where Icheon is located, historically was very similar to export from Icheon City, implying other fabs located in Gyeonggi Province, mainly Samsung's Hwaseong and Pyeongtaek, did not directly export HBM to Taiwan and Malaysia. Given this finding, we believe we can switch to using export data Gyeonggi as a substitute for Icheon and our analysis should remain valid. But this will require closer monitoring going forward to see if there is any change to exports from other fabs in Gyeonggi.
- In contrast to Samsung, export from N. Chungcheong & Gyeonggi, our proxy for hynix, remained weak in Aug. Export was up 7% MoM but still down 21% QoQ (vs May) ([Exhibit 7](#)). Jul+Aug export was down 21% QoQ as well, which contrasts with our expectation of QoQ increase in HBM revenue in 3Q26 ([Exhibit 8](#)). Regression still finds only US\$5.9B HBM revenue by SK hynix in 3Q26, ~14% below 2Q26 and 52% below our forecast ([Exhibit 9](#)). SK hynix's monthly seasonality was much more balanced than Samsung's, meaning it is less likely to significantly rebound in Sep exports ([Exhibit 10](#)). Overall export data has been weak and trailed further behind Samsung's for Jul+Aug. The export data won't perfectly capture hynix's HBM revenue and the predicted 3Q26 revenue can be inaccurate. But historically they have shared good correlation and warrants investors' attention. We think the weak export data is perhaps related to the delayed HBM4 shipment which SK hynix also alluded to in its earnings (see [takeaway](#)). So we think it is still possible to see more meaningful recovery later this quarter. We also note if we apply the most backend loaded seasonality (2Q24) since 2024, SK hynix's 3Q26 HBM revenue may reach US \$7.5B, which would still be growing from 2Q26. All things considered, data continued to suggest slow HBM momentum at SK hynix and that its market share likely will drop below Samsung's.

EXHIBIT 1: Being a close proxy for HBM shipment, Korea multichip memory export to Taiwan and Malaysia was solid in Aug, rising 17% QoQ.



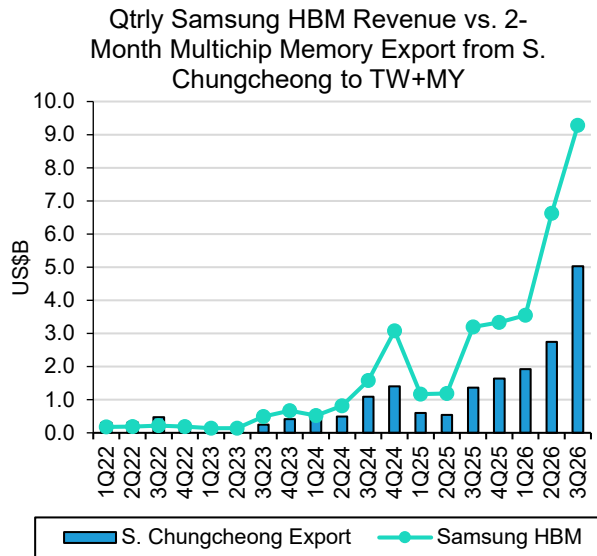
Source: Korea Custom Service, KITA, company reports, Bernstein estimates and analysis

EXHIBIT 2: Export from S. Chungcheong grew 61% QoQ.



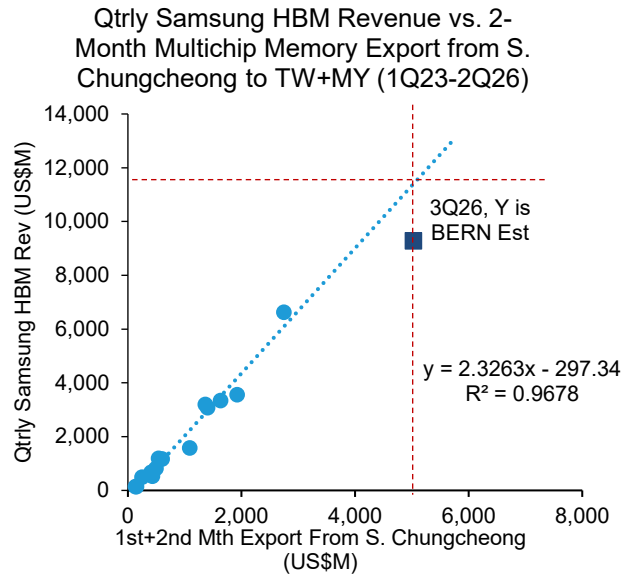
Samsung HBM revenue in 3Q26 is Bernstein estimate

Source: Korea Custom Service, KITA, company reports, Bernstein estimates and analysis

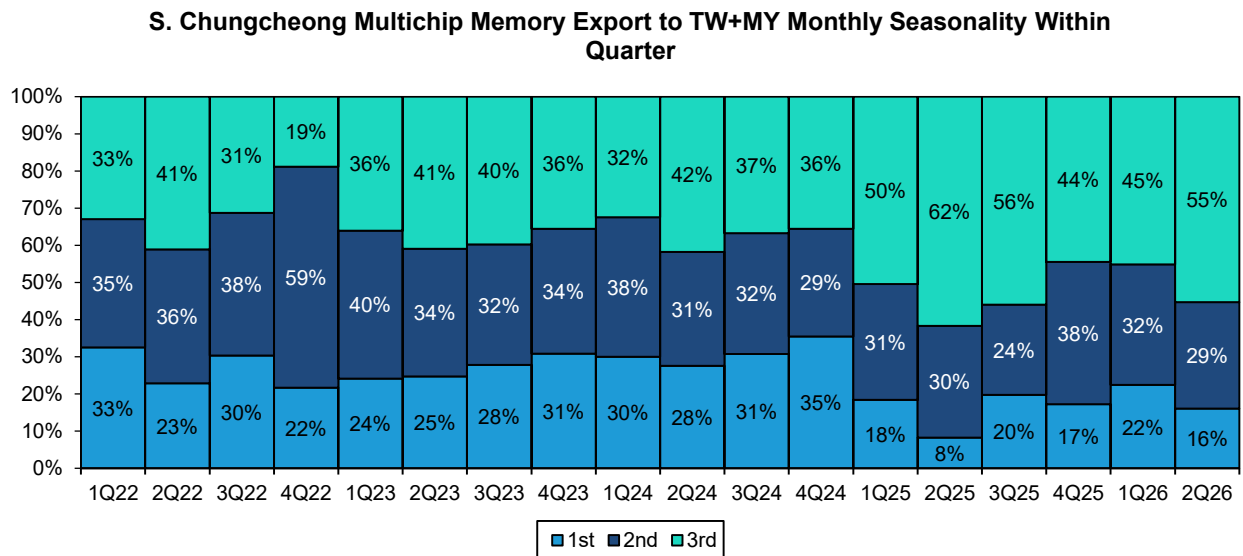
EXHIBIT 3: Jul+Aug export data suggested significant growth for Samsung 3Q26 HBM revenue.

Samsung HBM revenue in 3Q26 is Bernstein estimates

Source: Korea Custom Service, KITA, company reports and Bernstein analysis

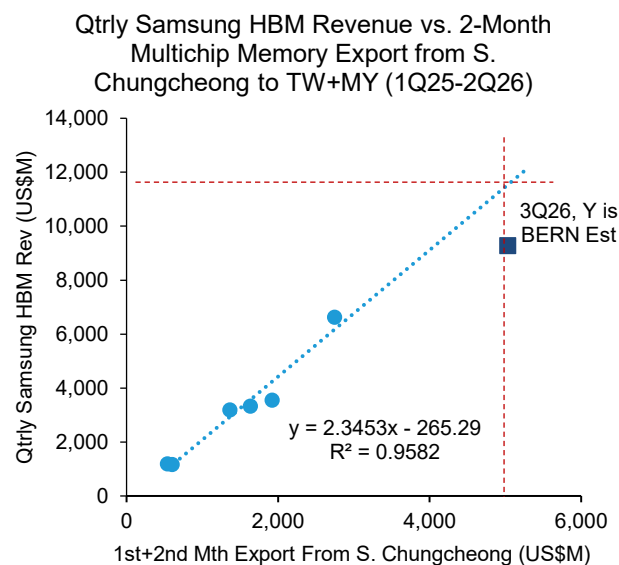
EXHIBIT 4: Per regression based on historical data, Aug data suggested \$11.4B HBM revenue for Samsung in 3Q26, up 72% QoQ.

Source: Korea Custom Service, KITA, company reports and Bernstein analysis

EXHIBIT 5: Monthly seasonality of export from S. Chungcheong has become more back-end loaded, suggesting Dec may see a strong increase in export.

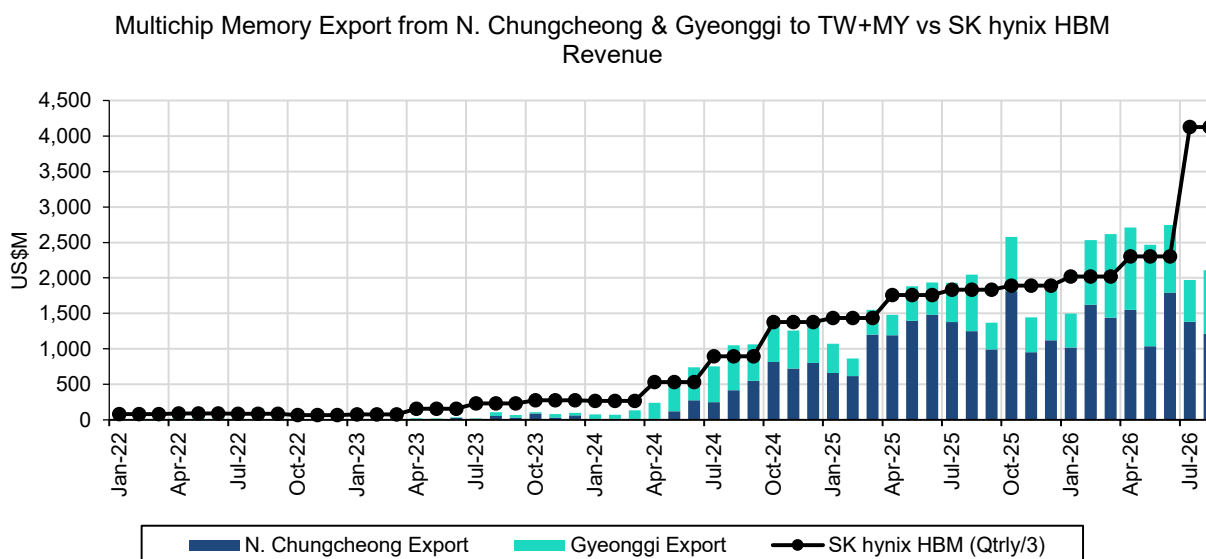
Source: Korea Custom Service, KITA and Bernstein analysis

EXHIBIT 6: Conclusion is largely the same if we use only the data from recent quarters.



Source: Korea Custom Service, KITA, company reports and Bernstein estimates and analysis

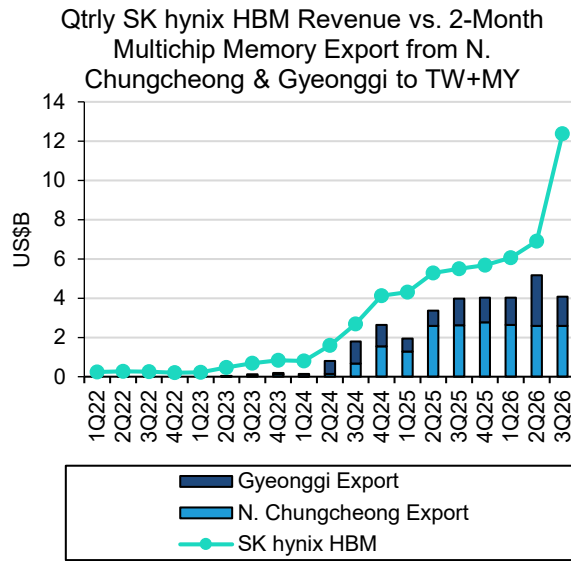
EXHIBIT 7: Export from N. Chungcheong and Icheon was up MoM in Dec.



SK hynix HBM revenue in 4Q25 is Bernstein estimate

Source: Korea Custom Service, KITA, company reports, Bernstein estimates and analysis

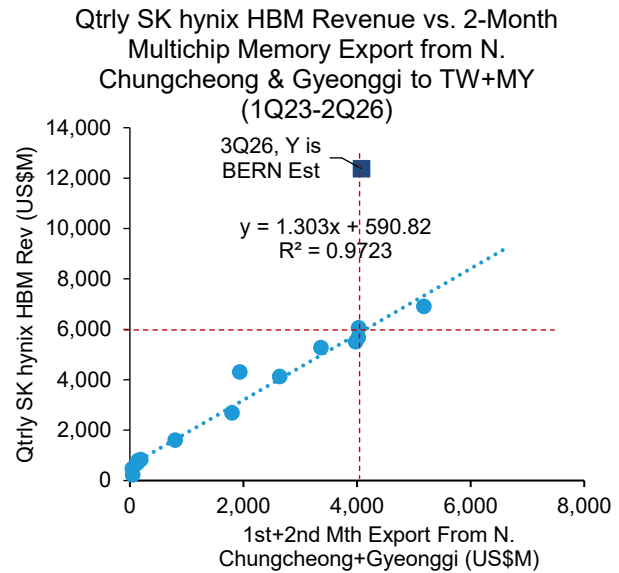
EXHIBIT 8: **Jul+Aug export data did not suggest SK hynix's HBM revenue will grow QoQ in 3Q26.**



SK hynix HBM revenue in 3Q26 is Bernstein estimate

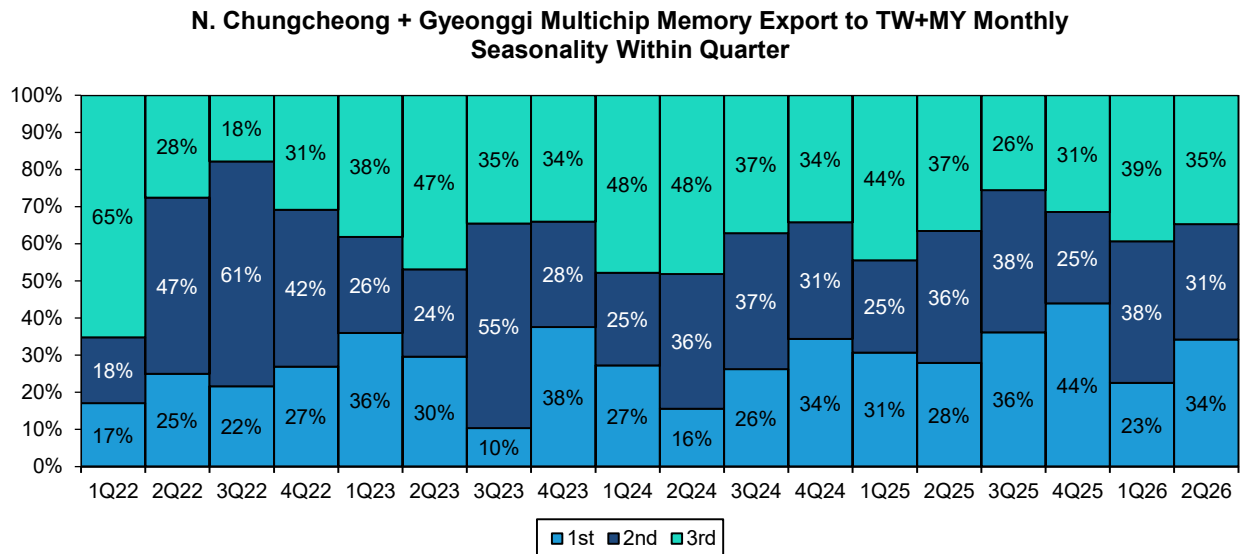
Source: Korea Custom Service, KITA, company reports and Bernstein analysis

EXHIBIT 9: **Regression suggests US\$5.9B HBM revenue for SK hynix in 3Q26, ~14% lower QoQ.**



Source: Korea Custom Service, KITA, company reports and Bernstein analysis

EXHIBIT 10: **Monthly seasonality for N. Chungcheong + Ichoen hasn't shown a notable sign of back-end skew and hence does not suggest that Dec will be strong.**



Source: Korea Custom Service, KITA and Bernstein analysis

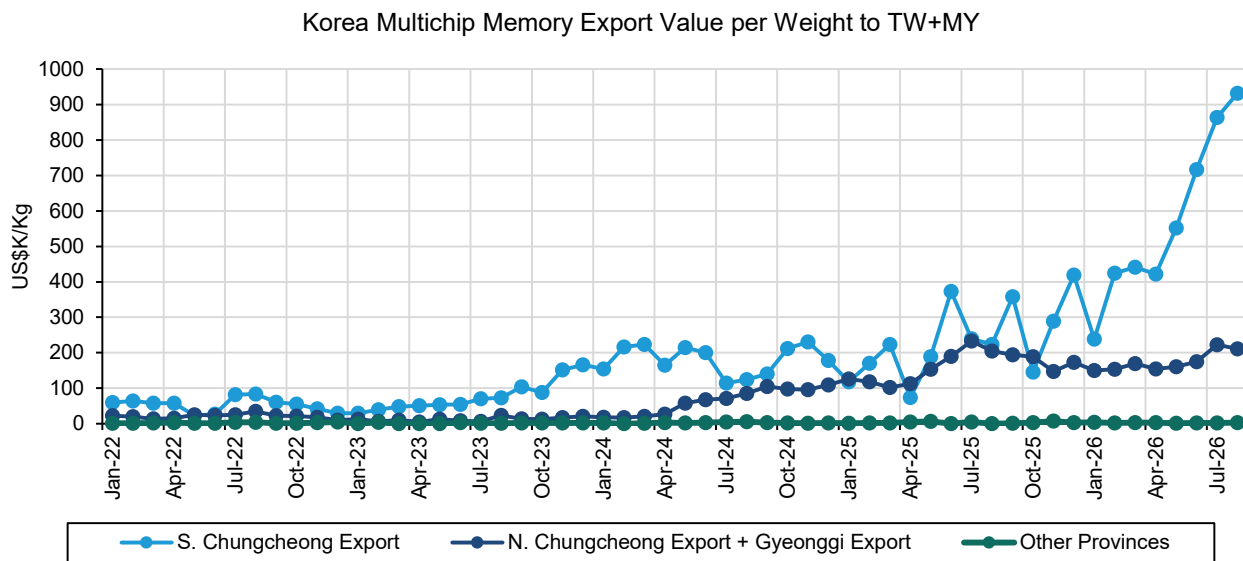
Samsung's export value per weight continued rising in Aug, indicating a continued increase in HBM4 shipment.

- Export value per weight for Korea multichip memory loosely correlates with HBM ASP, though it likely is only directionally indicative and not significant enough to make precise projections. For Aug, export value per weight was flattish MoM for N. Chungcheong+Gyeonggi (proxy for SK hynix). On the other hand, data for S. Chungcheong (proxy for Samsung) remained on a tear, rising by another 8% MoM. It was 2.2x higher over Apr level and over 5x over SK hynix's (Exhibit 11). This likely was most driven by the rising shipment of HBM4, which should command a notably higher value per weight than HBM3/3E. Samsung guided its HBM4 sales to >3x QoQ in 3Q26 and to account for well over 60% of its total HBM sales in 2H26, and

the export data has been well supporting the guidance.

- For SK hynix, data still suggested no sign of meaningful HBM4 shipment in the mix, in line with the weak export value data as well. But we should see value per weight rising more notably in coming months once SK hynix HBM4 shipment begins.
- Another important observation is while the price of conventional memory has increased ~5x since 3Q25, the data suggested that of HBM has been much steadier. Export value per weight from S. Chungcheong has been better, but it may be thanks to Samsung's earlier HBM4 shipment & hence a mix improvement, rather than a like-for-like HBM price hike. And the improvement vs. the prior level thus far has been only ~2x, still significantly below the size of the conventional DRAM price hike. The blended average might have even declined YoY when we consider SK hynix's larger scale in HBM and the falling value per weight trend for the company. That said, we believe that suppliers and customers are negotiating the HBM contracts and the prices for 2027 now, and the benefit of higher HBM prices will become visible later.

EXHIBIT 11: Value per weight data suggested Samsung is improving its HBM ASP quickly but not for SK hynix.



Source: Korea Custom Service, KITA and Bernstein analysis

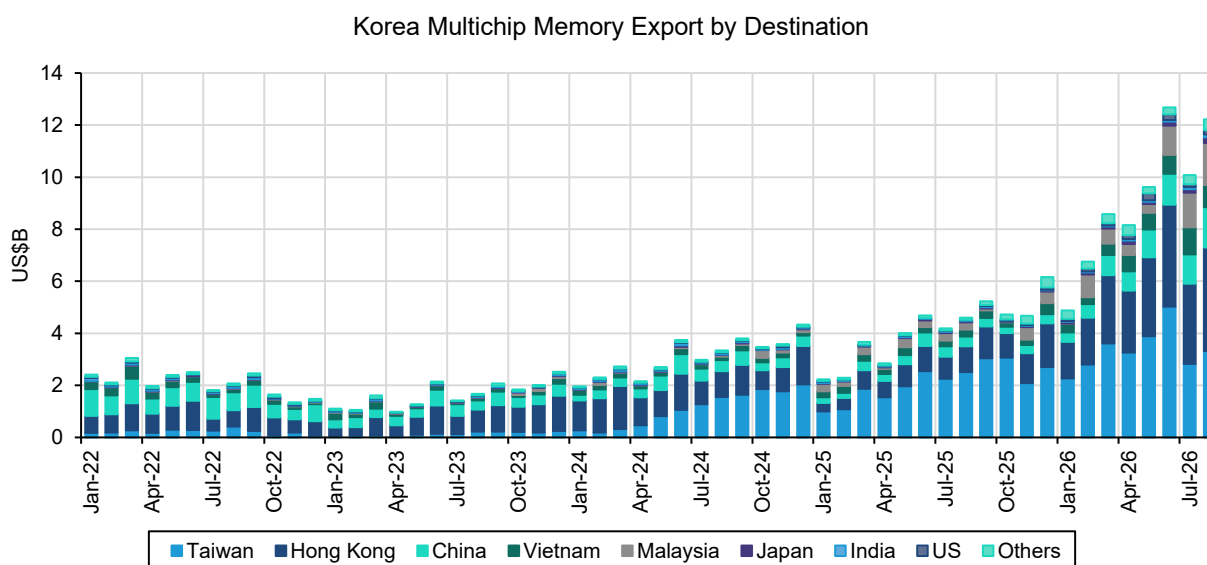
Export to Malaysia grew steadily to 33% of the total HBM exports in Aug. It is likely for Intel EMIB, but we wonder why the increase happened so early. Besides that we found no sign of new HBM packaging sites in Korea or new export destination countries.

- Korea's total multichip memory export to all countries was US\$12.2B in Aug, rebounding by 21% MoM on seasonality and also up 27% QoQ, thanks to both growth in HBM and also surging prices of conventional memory. Taiwan and Malaysia accounted for 40% of the total export, down QoQ from 44% in May, as price increase made conventional memory export value to other destinations rise more ([Exhibit 12](#)). Taiwan and Malaysia continued to lead other destinations by a wide margin on export value per weight, and hence we continued to include only Taiwan and Malaysia as the primary destination of HBM export from Korea ([Exhibit 13](#)). Export value per weight to Hong Kong and China Mainland was broadly still rising too. Additionally, we also observe a notable increase in other regions such as the US, Japan and India. Thus we still assume the increases in these regions were mostly due to rising conventional memory prices, instead of HBM shipment, as Samsung and SK hynix likely exported considerable amounts of mobile memory packages to these regions & memory packages are categorized as multichip memory too ([Exhibit 14](#)). And it also supports conventional memory enjoys rapidly rising prices and commands better revenue and profit per wafer than HBM now.
- Export to Malaysia rose by another 22% MoM to ~US\$1.6B in Aug. Average run rate in the last 6 months was to US\$0.9B, compared to US\$3.7B for exports to Taiwan. We understand that Malaysia is home to Intel's high-end packaging facilities, where some high-end server CPUs equipped with HBM are packaged with EMIB. Separately we picked up a growing customer interest in EMIB, and we note Intel is set to bring online a new advanced packaging fab into production this year as well ([link](#)). However, even if customers decide to adopt, we believe any volume production should be slightly over 1 year

from now, and wonder why HBM export to Malaysia took place so early. All in all the export to Malaysia is worth continued monitoring.

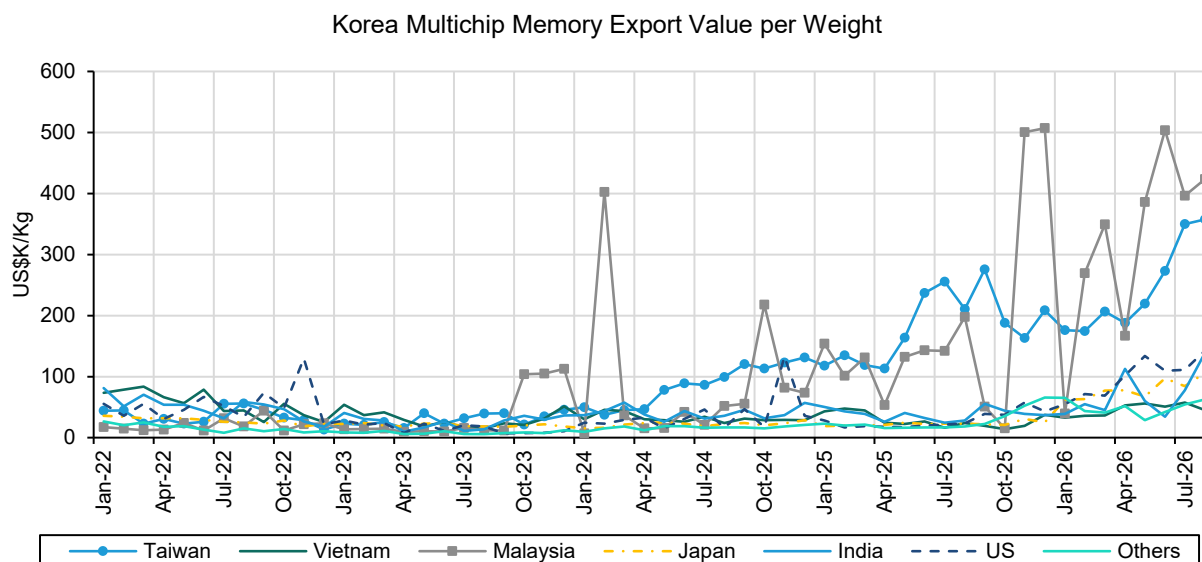
- Export to Malaysia was still mainly coming from S. Chungcheong, where Samsung's backend fabs are (Exhibit 15). Shipment from N. Chungcheong and Gyeonggi, proxy for SK hynix, also rose to ~30% of all export to Malaysia in Aug. Samsung likely benefits more from the Malaysia demand, with ~40% of its export going to Malaysia, while Malaysia was only 24% of the export of Chungcheong and Gyeonggi (proxy for SK hynix) (Exhibit 16). Broadly the export to Malaysia has seen meaningful growth this year and has contributed to HBM growth especially at Samsung. It also supports Samsung's broadening customer recognition in HBM.
- For the origin of export, Aug data showed that S. Chungcheong, N. Chungcheong and Gyeonggi continued to account for almost all of multichip memory export to Taiwan & Malaysia (Exhibit 17). So it does not appear either Samsung or SK hynix has started packaging HBM in other locations within Korea.
- Overall, all the data above suggested that HBM export continued to concentrate in mostly Taiwan and to a lesser degree Malaysia. However, we'd like to point out the limitations to our methodology. One is if either of them moves any HBM packaging outside of Korea, it will not be picked up by this export data. We however don't think this is happening soon. Secondly, if any of their HBM shipment is consumed (i.e. packaged into another product) within Korea, it will also not be reflected in our analysis either. We think this second limitation may become more relevant as Amkor (AMKR US, not covered) has small volume of CoWoS-like production in Korea already and may increasingly do more in the future.

EXHIBIT 12: Multichip memory includes mobile DRAM & NAND packages too, and most of the export went to Hong Kong/China & Vietnam prior to 2024 as most of smartphone production is in those regions. Since 2024 the export to Taiwan has been rising very quickly due to a surge in HBM export to Taiwan for CoWoS packaging there. And in recent quarters we also saw an increase in the export to Malaysia.



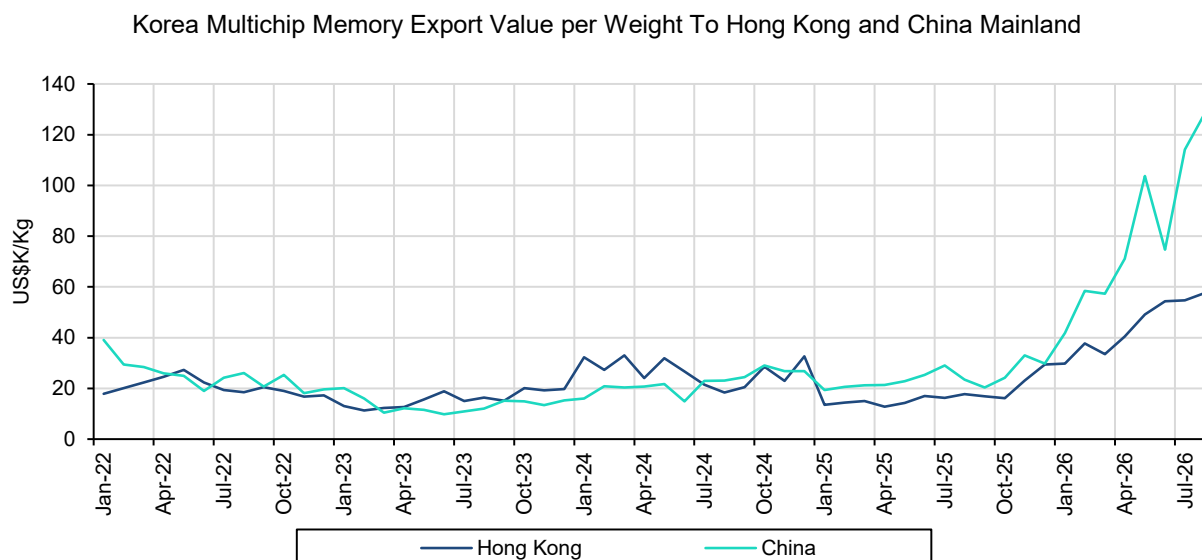
Source: Korea Custom Service, KITA and Bernstein analysis

EXHIBIT 13: **Value per weight continued to indicate that HBM was likely primarily exported to Taiwan and Malaysia in Aug.**

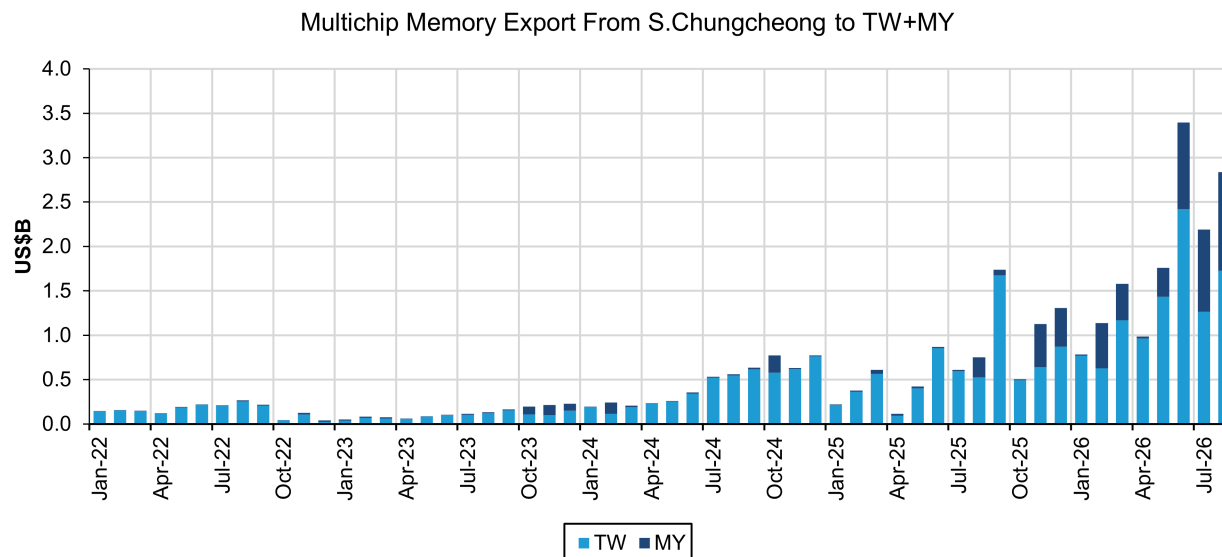


Source: Korea Custom Service, KITA and Bernstein analysis

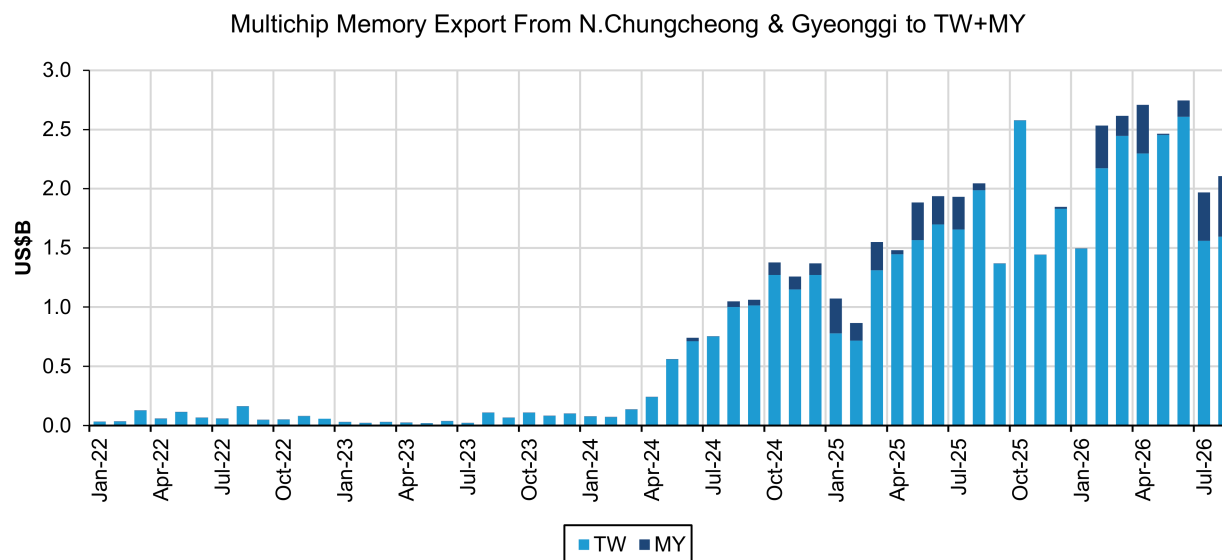
EXHIBIT 14: **Value per weight generally has been growing for Hong Kong and China. We assume the changes were due to conventional memory increase instead of HBM shipment.**



Source: Korea Custom Service, KITA and Bernstein analysis

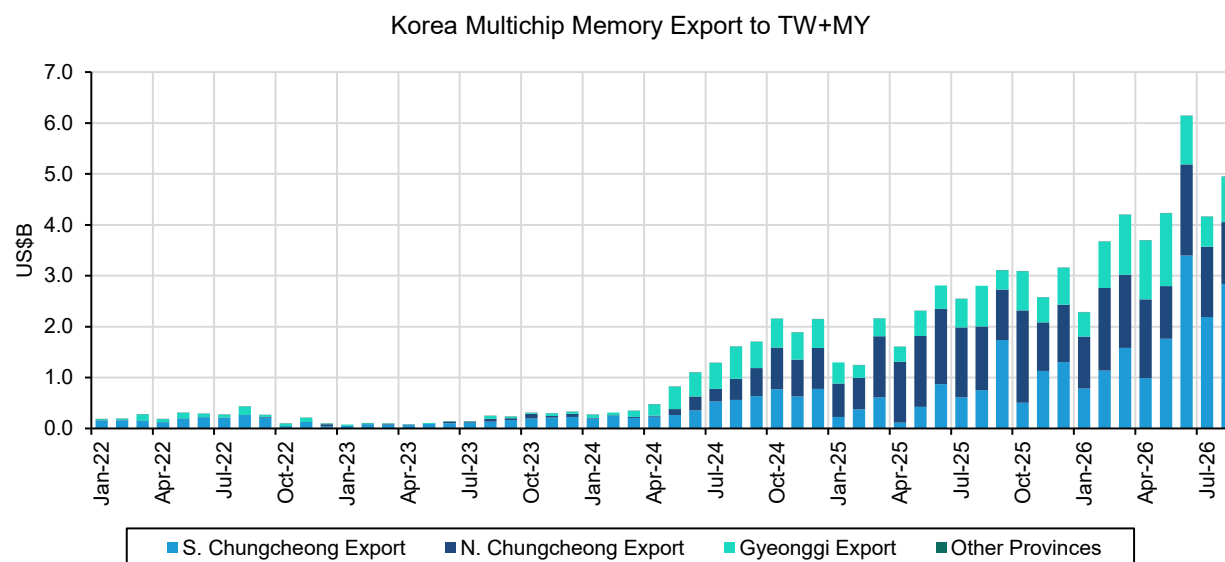
EXHIBIT 15: **Samsung HBM export to Malaysia grew steadily in Aug.**

Source: Korea Custom Service, KITA and Bernstein analysis

EXHIBIT 16: **Export data suggested SK hynix may be also exporting more HBM to Malaysia in Aug, though size remained much smaller vs Samsung's.**

Source: Korea Custom Service, KITA and Bernstein analysis

EXHIBIT 17: S. Chungcheong, N. Chungcheong and Gyeonggi continued to account for almost all of multichip memory export to Taiwan & Malaysia.



Source: Korea Custom Service, KITA and Bernstein analysis

Overall Aug continued to indicate strong HBM growth momentum at Samsung, but slow progress at SK hynix.

- In summary, Aug export data continued to support strong growth for Samsung HBM revenue in 3Q26, but much weaker momentum for SK hynix. Regression suggests 72% QoQ growth in HBM revenue at Samsung in 3Q26, and likely a notable ASP increase too driven by HBM4 ramp. For SK hynix, export data suggested 14% QoQ drop in HBM revenue in 3Q26. We note that prediction based on export data can be flawed and inaccurate, but given good historical correlation, still worth noting to investors. Export to Malaysia grew even higher to reach 33% of all HBM export in Aug.
- For the stocks, price increases in conventional memory, more clarity on capital return & LTAs will likely provide support to share prices. Mid- to long- term, we will be monitoring progress at CXMT & also IPO of YMTC to better assess their structural threat, especially in NAND as China can advance technologies and capacity of NAND without EUV. We caution investors of cyclical risks but remain constructive on DRAM and on Samsung, SK hynix and Micron (see [Global Memory: Memory becoming a burden of AI too?](#)), especially after the recent pullback. For KIOXIA, despite the substantial correction and more aggressive shareholder returns, we remain cautious due to structural concerns on competitive dynamics in NAND. And we find current stock price still somewhat expensive, based on our detailed SOTP valuation ([KIOXIA: All parties come to an end - Underperform](#)).

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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VALUATION METHODOLOGY**Samsung Electronics Co Ltd**

We value Samsung Electronics with 6.2x of forward 5Q-8Q EPS of KRW71,172 and arrive at price target of KRW 440,000 for its common shares, KRW 374,000 for preferred shares, and US\$7,350 for GDR.

SK Hynix Inc

We value SK hynix with 6.2x of forward 5Q-8Q EPS of KRW536,158 and arrive at price target of KRW 3,300,000.

Micron Technology Inc

We value Micron with 7.7x of forward 5Q-8Q EPS of US\$169.8 and arrive at price target of US\$1,300.

KIOXIA Holdings Corp

We value KIOXIA with 4.1x of 1-year forward EPS of ¥9,810 and arrive at price target of ¥40,000.

RISKS**Samsung Electronics Co Ltd**

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

SK Hynix Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

Micron Technology Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

KIOXIA Holdings Corp

Upside risk to our target price include 1) stronger than expected NAND demand growth, which likely would lead to better NAND pricing and hence company earnings, 2) favorable policy support from Japanese government such as capex subsidies, and 3) better than expected cost reduction

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

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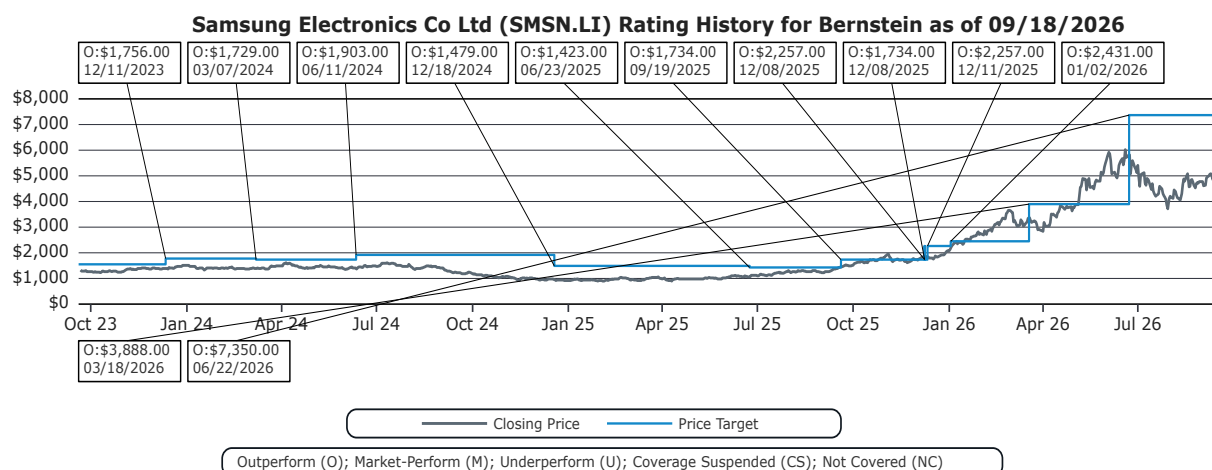
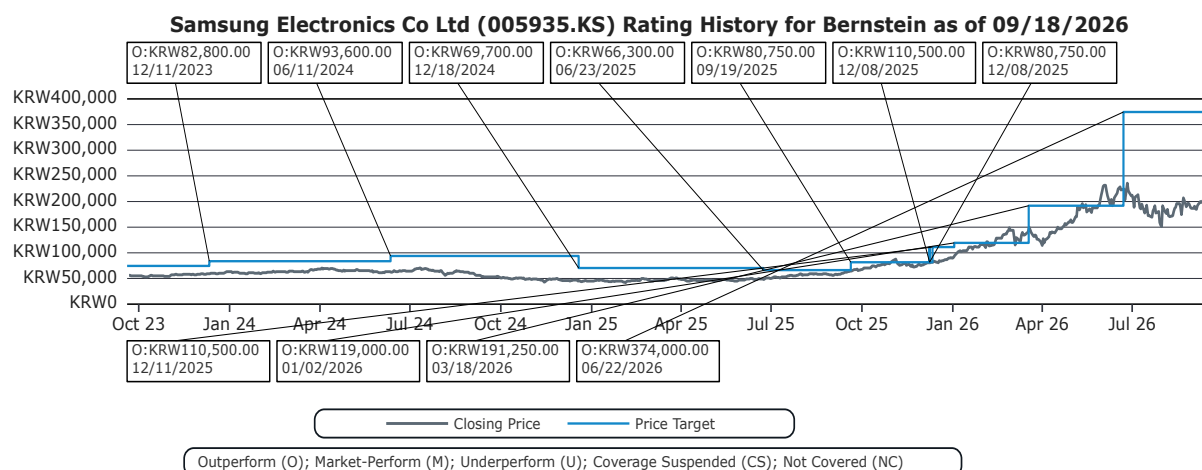
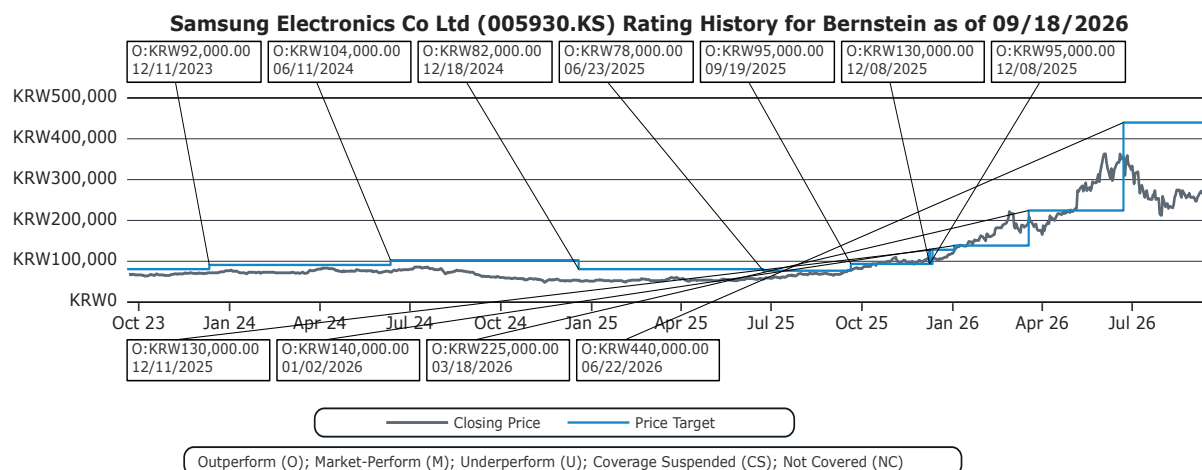
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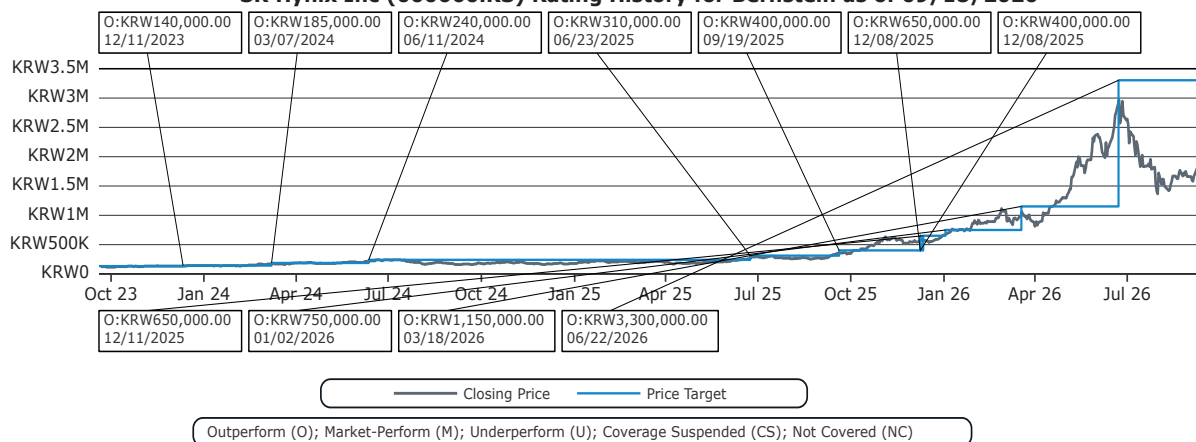
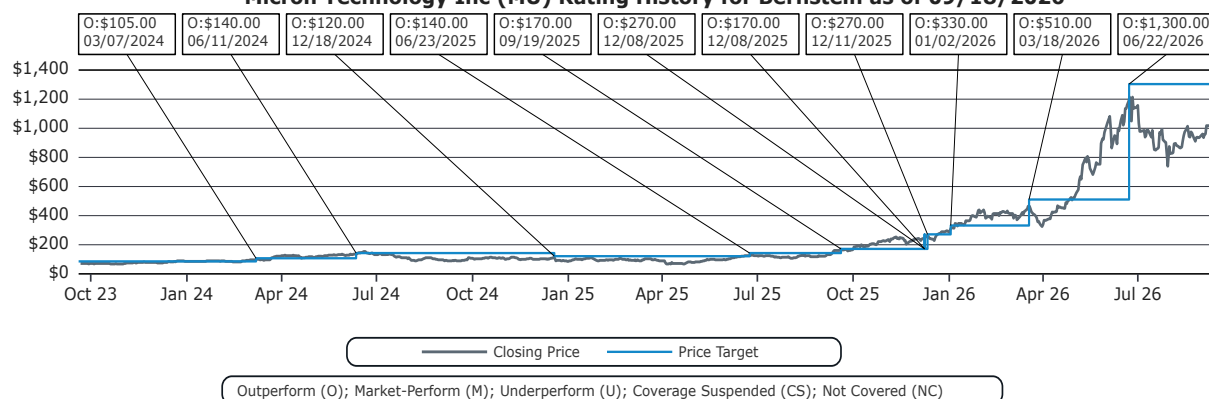
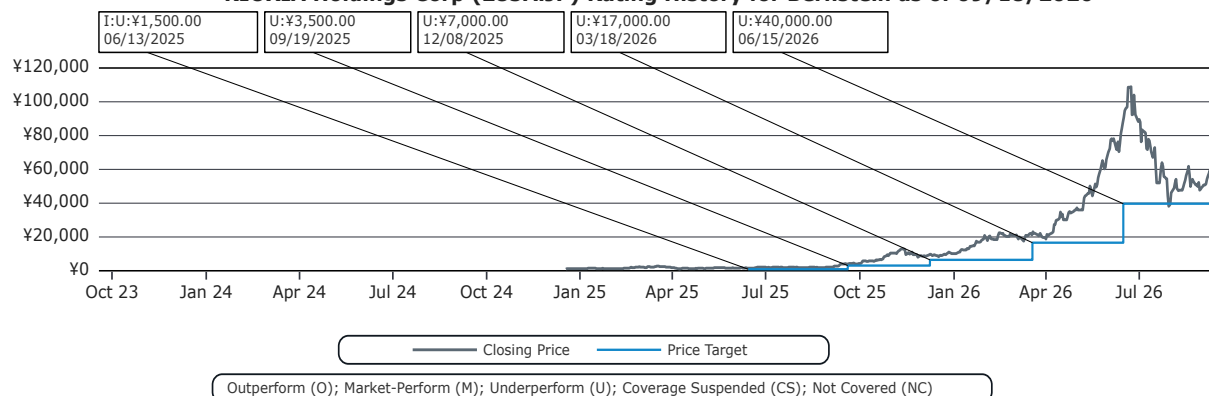
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